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Bird Sound Identification System using Deep Learning

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Abstract

This paper identifies the model for classifying the bird sounds using the Convolutional Neural Network (CNN) algorithm. CNN are considered as one of the powerful toolkits developed in the machine learning and that have shown to be more efficient particularly in the field of image processing and sound recognition. The two-step verification process which will participating in the creation of datasets and neural network input. This system uses the collection of labelled and categorized recordings. Spectrograms are generated from the downloaded data which is used as the input for the neural network. The experiments are conducted and it will match with the different configurations, which including the number of bird species and Spectrograms. As a result, we can identify the bird species from its audio recordings. By identifying the bird species on a large scale, it provides a vital information about our ecosystems and biodiversity. The changes in the ecological environment can be reflected by the quantity and composition of bird species. It is one of the major challenges which leads to recognition of many bird species based on their audio. In modern years, the deep learning techniques namely convolutional neural networks (CNN) have played a major role in many areas of science and also successfully provided efficient methodologies, such as pattern recognition, object detection in images, and classification of patterns in audios and speech recognition.

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Keywords: **Co**nvolutional Neural Network; Bird Sound Recognition; Transfer Learning; Audio Classification

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1. Introduction

1.1 Overview

In this system, predefined audio recordings of bird calls are used to identify the various bird species.

1.2 Purpose

Birds are considered as one of the nature's attractive creatures and it is also an important existence of our ecosphere. They are indicators of biome quality and thus play an important role in carrying of many environs functions and it particularly act as a regulator for the insect population as insectivores, dispersal agent for plant seeds and majorly in the flower pollination [7].

Bird-habitat and classification is the process of examining and identifying birds in their nature habitats by their sound. It is also a provides a useful way for the environmentalists, wildlife researchers and ornithologists. The uncontrollable human as well as natural activities such as habitat loss, overhunting, Air, Water and Soil contamination, competitiveness and pillaging from non-domestic invasive species and changes in the climate has led to threaten the bird population and it has globally increased the responsibility for the protection in addition with conservation of natural life and habitat of birds [2-11]. Birds are critical to functioning of the ecosphere. Therefore, it provides the techniques and ability which make the bird monitoring more efficiently and accurately. Apart from increasing efficiency it will also massively benefits to the conversation efforts[12-20]. In this system sounds of different bird species is identified by using Deep Learning concept[21-25]. And, the concept of algorithm used in this process is mentioned in below diagram - figure 1 Deep learning techniques.

1.3 Motivation

- To provide a model for classification of Bird sounds using Deep Learning concept.
- Its main aim is to identifying and classifying sounds as simple audible species.

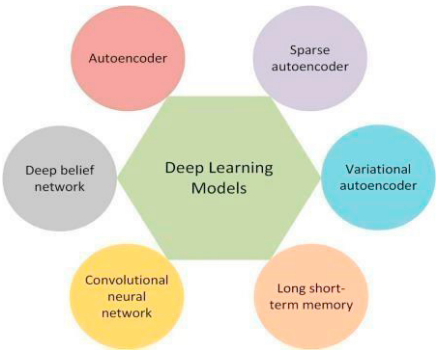


Figure 1. Deep Learning Technique

Deep Learning is a technique which is used in Machine Learning. It mainly involves in the modelling of neural networks to solve the complex problems. Neural networks are modelled on the basis of the structure and working function of the human brain. To process and transform data, neural networks contain various interconnected nodes.

2. Materials and Methods

2.1 Dataset - Recordings of Bird Sound

Xeno-canto is a community – driven database that collects the various sounds of birds from contributors among worldwide. It hosts a vast collection of bird recordings, with over 4,00,000 recordings from the thousands of species. The data which is available in various formats for the downloading purpose, including MP3 and WAV.

The bird songs can be identified by using this dataset and it contains the audio recordings of bird sounds collected from different sources, including Xeno-cento. It includes various bird species and provides annotation for species labels. The Index of the files contained in xeno-canto and it ranges from 0 to 2,729 data collections.

This dataset is based on the Cornell Lab of Ornithology's Birdsong Recognition competition on Kaggle. It includes audio recordings of various Bird species along with associated metadata such as species labels, location and recording details.

It contains a recordings subset from the California and Nevada, United States of America which including a balanced amount of sample collection (30) per species. Recordings are labelled by species and contain unstandardized labels for call type. It composing of both male and female songs, recorded in different habitats[13].

Whereas the dataset which contains the 91unique values of various birds such as Abert Towhee, Acorn Woodpecker, American Crow, Dusky flycatcher, Anna's Hummingbird, Bell's sparrow, Black Phoebe, Cassian vireo, common poorwill, Elegant Tern, etc... are the well-known bird species collections. Finally, the genus of labelled species can be categorized as Vireo 5%, Empidonax 3 % and other 2,490 species are in the range of 91%.

2.2 Proposed Method

The proposed system can be able to automate the sound processing and Convolutional Neural Network (CNN) techniques which is primarily involved in the process of identifying the bird calls. Additionally, this system uses LSTM method for the effective feature identification.

The first step involved in this system is to create a database which consists of recordings of the bird species in mp3 format. These sound recordings are then composed to the preprocessing methods. Spectrograms which are generated from those audio clips and thus it is given as an input to the CNN. Then the fragmentation is carried out in the audio for the purpose of noise removal. The identification process can compose of Evaluation and Training phases[1]. Such system has been developed by utilizing the trained dataset which consists of trained and test results by using those results we can able to predict the various birds with its accuracy.

2.3 Proposed system framework

The intensity of Bird call can vary between different sound samples. Therefore, it is important to normalize the dataset such that the intensity values are within the same range. By adding more layers which lead to improve accuracy but this may also lead to affect the training and influence time as increase in the number of trainable parameters and computational cost. We resize the images and test our CNNs in order to explore how the input shape affects models with a different number of Convolutional layers. The issue of over training can be handled by using Regularization techniques which reduce generalization error. Convolutional layer is frequently preceded by pooling layer. These layer down sample the map feature with the use of a filter and eliminating irrelevant features that can be considered as noise. Choose Max pooling it is extensively used in CNN and is more effective in simple classifier. Cross validation has to be performed and it carries out the unbiased estimate of the generalization error. Next the kernel adjustment would be changing the number of kernels per layer. Finally, the CNN model is used to train and test the data, as mentioned in the below diagram i.e. (Figure 2).

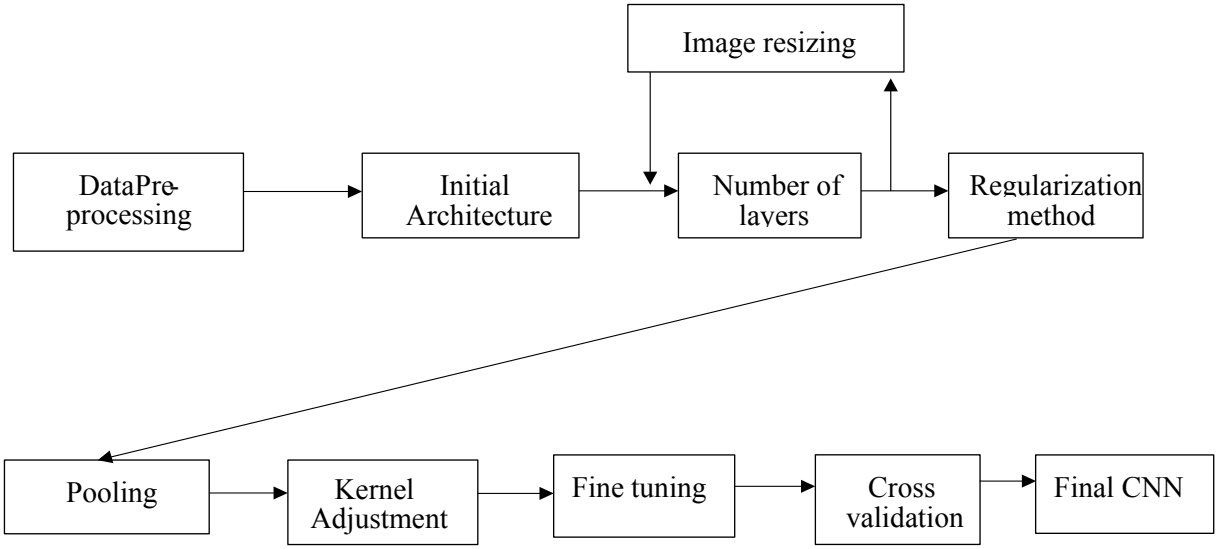


Figure 2. Proposed System Framework

Therefore, the proposed system approach involves in the comparative study of both CNN and LSTM models.

2.4 Setup and tools

For this study, we used Python 3.7 version, and the Python libraries which is used to develop the system consists of Tensor Flow lite, Keras, NumPy, Scikit learn, to create and train our models, we utilised TensorFlow version 2.10 and Keras version 2.10.0. And also, we conducted the training on Google Colab, which uses similar versions of TensorFlow and Keras frameworks as our local environment.

3. Methods and Architecture

This developed system primarily uses the LSTM and CNN approach and it can be stated as below mentioned ways

3.1 LSTM

LSTM stands for Long Short-Term Memory networks it is used in the field of deep learning. It can also use to learn, process, and classify sequential data because it can be involved in long term dependencies between time steps of data. In this system, LSTM can able to use one feature at a time, so each feature is fed into the LSTM one by one and the best features is chosen according to this and for ensemble classification of several bird species.

3.2 CNN

A CNN is a type of Deep learning algorithm and it is a Multilayer perceptron that is applied to the process of analysis and learning of visual features from large amount of data. It is also helps in extracting the features for Classification and detection.
In this system, Several CNN model Architecture were compared including two Max pooling Convolutional 2 D

repeating layers model and an encoder –decoder model. CNN typically consists of several types of layers, as mentioned in below diagram i.e. (Figure 3)

Input layer

The input layer is the initial layer and where the input data is received in this layer. Each node in the input layer can able to represents a feature of the input data and it determines the dimension of the input data. From this raw input data, the neural network can have the capability to extract the important features through the hidden layers during the process of training.

Hidden layer

It is an intermediate layer between the input and output layer. Deep networks consist of multiple hidden layers, and it help us to learn complex patterns in the data. Hidden layer consists of convolutional layer followed by other layers such as pooling layer, normalization layer and Fully Connected layer. Each node has the ability to perform a mathematical operation on the input and finally passes the result to the next layer.

Output layer

The output layer is the final year and it produces the output of the deep neural network. The number of nodes which is depends on the type of task and design of network. Before the derivation of the final output the weight and biases are applied to get the desired final prediction.
The commonly used activation functions in output layer are

- Linear
- Sigmoid
- Softmax

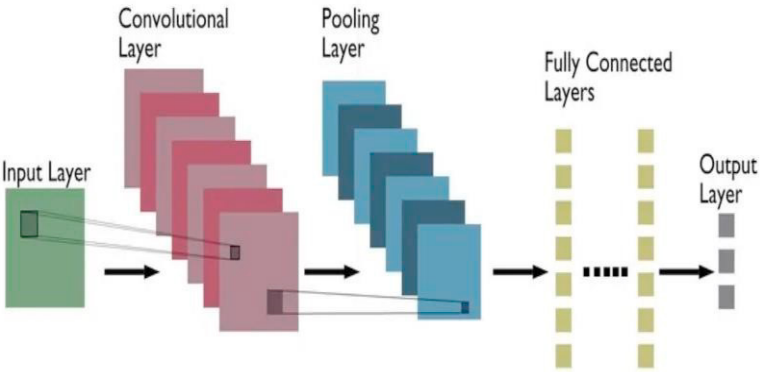


Figure 3. CNN Architecture

3.3 Model Representation

There are two types of Keras Models – Sequential Model and Functional API. In which Sequential Model is basically a linear composition of Keras Layers. It is easier and simpler. Thus, it also provides a minimal way to represent nearly all available neural networks. The Model class is used to create customized models as well. The creation of own complex model can be done by using sub-classing concept.
The Functional API is basically used to create complex models. In Keras model each layer is represented as following input layer, hidden layer and output layer in actual proposed neural network. To create any complex neural network, the pre-build layers of Keras model can be used.

4. Result

This part of the paper presents and discusses the results that were obtained after conducting the experiments in our paper shown figure in 4 and 5 The technical novelty of this work will be given to achieve desired results

LSTM

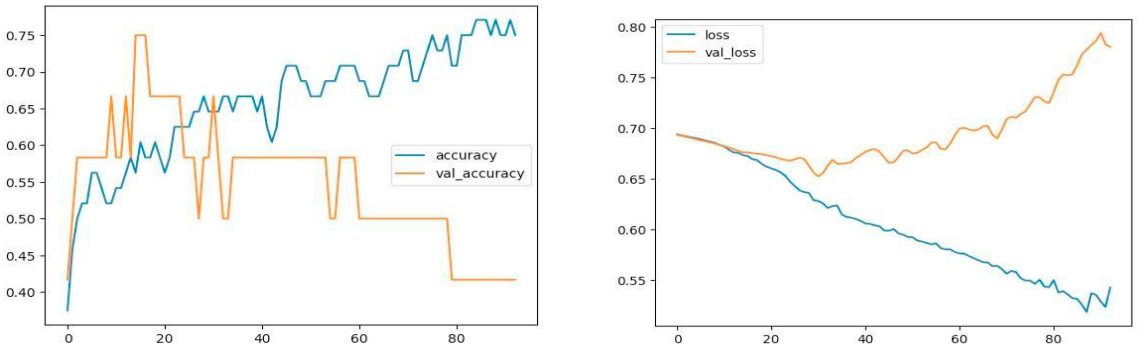


Figure 4. Performance Measures LSTM

CNN

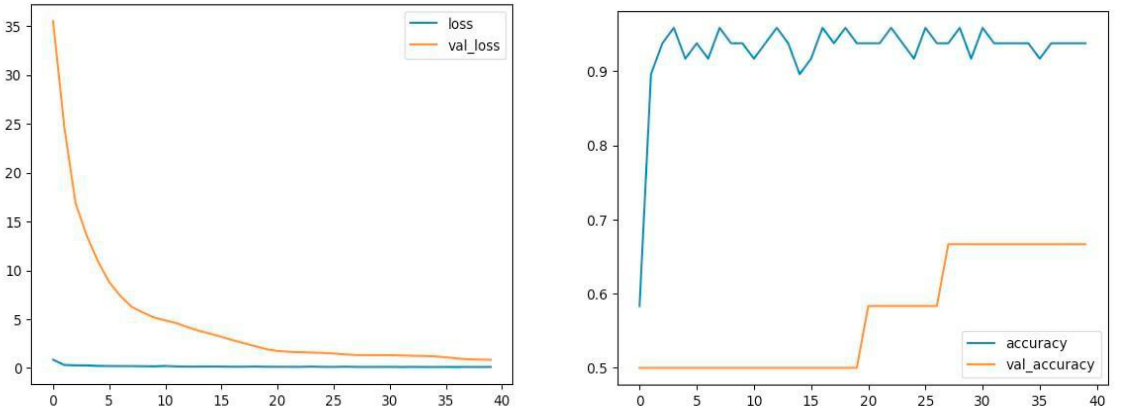


Figure 5. Performance Measures CNN

Visualizing the accuracy vs. validation accuracy over a few epochs is a best way to determine if the model has been sufficiently trained. The values can be optimized by visualization of the fitness of the training and validated set of data which helps to build a better model. Matplotlib library is used to visualize the output.

5. Conclusion

The main motive to develop the system is to identify the bird creatures from their calls in environment which is a major challenge. This study aims at exploring the major use of deep learning techniques CNN and LSTM in Bird Sound Identification System. Both LSTM and CNN Classification methods are used to predict the specific bird sound.

It is easier to manage the system by recording the sounds of birds, by using which the corresponding species of birds can be known. The Bird species can be known by recording the sounds of the bird, and it will make the system easier to manage. In this work MFCC, has been used and tested through different algorithms. And, after the Pre-processing process, we generated a spectrogram and used it to train the model for classification. The model can be trained by using the data of real bird sounds that was recorded in various environment and natural habitats. Results were observed at various values of amplitude, magnitude, number of epochs, and data partitioning. The results of exploration in this study are not limited to the following, the PCA used on deep features reduce the dimensionality as well as it reduced the training and testing time but it does not allow to increase the identification accuracy LSTM and CNN classifier.

The system has provided 75% accuracy in prediction of finding bird species. It is recommended that to conduct more investigation to improve accuracy and to reduce the training time using different algorithms. This can improve the accuracy by fine -tuning the performance parameters.

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